최웅 (Woong Choi)

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Date of birth	29 January 1997

Education

2022.03.01- 2028.02 (expected)	**Ph.D.** Department of Chemical and Biological Engineering, Seoul National University, Seoul, Republic of Korea Advisor: Prof. Jihyun Kim
2016.03.01- 2022.02.25	**B.S.** Department of Chemical and Biological Engineering, Korea University, Seoul, Republic of Korea

Projects

2022.04.- present	극한환경용 산화갈륨 자외선 센서 기술 개발, 한국산업기술평가관리원
2022.12- present	High NA 향 박막 재료 별 Chemical Etch Rate 특성 연구, 삼성전자 반도체연구소/Mask 개발팀

Awards

[1] 학생우수발표, 제6회 한국산화갈륨기술연구회 전문학술워크숍, 여수 (09.06-08, 2023)

Publications

[1] **W. Choi**, D.-W. Jeon, J.-H. Park, D. Lee, S. Lee, K. H. Baik and J. Kim, "Photo-enhanced metal-assisted chemical etching of α-gallium oxide grown by halide vapor-phase epitaxy on a sapphire substrate and its applications", *Mater. Adv.*, **4**, 4520 (2023)

[2] J. Park[1], **W. Choi[1]** and J. Kim, "Wet cleaning of Ta-based extreme ultraviolet photomasks at room temperature", *Nanotechnology*, **35**, 205701 (2024)

[3] S. Lee, J. Kim, **W. Choi** and J. Kim, "WSe_2/β-Ga_2O_3 p-n Heterojunction-Based Normally Off Phototransistors for Self-Powered UV-C Detection", *ACS Photonics*, **12**, 2112 (2025)

[4] J. Kim, H. W. Kim, **W. Choi**, J. Kim and D. Lee, "Annealing-free Ohmic contact of β-Ga_2O_3 *via* nitrogen plasma treatment", *J. Mater. Chem. C*, **13**, 10561 (2025)

[5] **W. Choi**[1], S. Lee[1], S. Moon, K. H. Baik and J. Kim, "Achieving Face-Selective Ohmic Contact to β-Ga_2O_3 *via* Anisotropic Trench Structure", *ACS Appl. Electron. Mater*. **7**, 8618 (2025)

[6] **W. Choi[1],** J. Park[1], M. Kim and J. Kim, "Improving the Particle Removal Efficiency of Sn Contaminants on High-NA EUV Masks via Photo-Assisted Wet cleaning", *ECS J. Solid State Sci. Technol.* **14**, 125002 (2025)

[7] Y. Lee, S. Moon, S. Ko, D. Lee, S. Yang, **W. Choi**, J. Kim and D. Lee, "Area-Selective UV Laser-Assisted Oxidation for Efficient Hold Injection in WSe_2 FETs and Its Application to Multifunctional Logic Operations", *ACS Appl. Electron. Mater*. (2026.06.23)

[8] J. Kim[1], **W. Choi[1]**, S. Lee, S. Moon and J. Kim, "Forward Metal-Assisted Chemical Etching for Self-Aligned Gate-Recess Enhancement-Mode β-Ga_2O_3 Metal−Semiconductor Field-Effect Transistors", *ACS Mater. Lett.* (accepted)

Patents

[1] 김지현, **최웅**, 전대우, 박지현, "촉매 습식식각을 이용한 산화갈륨 식각방법", Republic of Korea, 10-2022-0144719 (2022.11.02)

[2] 김지현, 이승윤, **최웅**, "포토 디텍터", Republic of Korea, 10-2025-0046552 (2025.04.10)

[3] 임슬기, 김우석, 오종근, 김문경, 김지현, 박제환, **최웅**, "포토 마스크의 세정 방법 및 장치", Republic of Korea, 10-2025-0098016, (2025.07.21)

[4] 김지호, 김지현, **최웅**, "반도체 소자", Republic of Korea, 10-2026-0039127, (2026.03.04)

[5] 임슬기, 김우석, 오종근, **최웅**, 박제환, 김지현, 김문경, "합금 박막의 습식 식각 방법 및 이를 이용한 EUV 마스크 패턴의 형성 방법", Republic of Korea, 10-2026-0048918, (2026.03.18)

Conference Talks

[1] Inverse Metal-Assisted Chemical Etching of β-Ga_2O_3 Using Quantitative Analysis, 제5회 한국 산화갈륨 기술연구회 전문학술워크숍, 대전 (08.30-09.01, 2022)

[2] 알파상 산화갈륨의 역-금속 촉매 습식 식각, 제30회 한국반도체학술대회, 정선 (02.13-15, 2023)

[3] Photo-Enhanced Inverse Metal-Assisted Chemical Etching of α-Ga_2O_3 grown on Al_2O_3, 243rd ECS Meeting, Boston (May 28-June 2, 2023)

[4] Photo-Enhanced Inverse Metal-Assisted Chemical Etching of Alpha-Phase Gallium Oxide, 제6회 한국산화갈륨기술연구회 전문학술워크숍, 여수 (09.06-08, 2023)

[5] β-Ga_2O_3/WSe_2 Anti-Ambipolar Transistor, ICAE 2023 7th International Conference on Advanced Electromaterials, 제주 (October 31-November 3, 2023)

[6] Anti-Ambipolar Transistor via β-Ga_2O_3/WSe_2 Mixed dimensional Heterostructure, 2024 KIChE Spring Meeting, 제주, (04.24 – 27, 2024)

[7] Plane Selective Metal-Assisted Chemical Etching for Good Ohmic Contact to β-Ga_2O_3, ECS PRiMe 2024, Honolulu (10.06 – 11, 2024)

[8] Applying plane contact in β-Ga_2O_3 via photo-enhanced metal-assisted chemical etching to improve

Ohmic contact, KOREA-JAPAN SYMPOSIUM ON MATERIALS AND INTERFACES, 서울, (11.27 – 30, 2024)

[9] Photo-assisted wet cleaning of Tin Contaminants on high-NA EUV Masks, SPIE Photomask Technology + EUV lithography, Monterey, (09.22 - 25, 2025)

[10] 광-보조 습식 세정을 통한 High-NA EUV 마스크 상의 Sn 오염 입자 제거 효율 향상, 제 33회 한국 반도체 학술대회, 정선, (01.27 – 30, 2026)

[11] Forward Metal-Assisted Chemical Etching for Self-Aligned Recess-Gate β-Ga_2O_3 MESFETs, 2026 MRS spring, Honolulu, (04.26 – 05.01, 2026)

[12] Forward-MAC etching of β-Ga_2O_3 and self aligned recess gate MESFET, Nano Korea 2026, 일산, (07.08 – 10, 2026)

Research Skills

Ⅰ Process integration for semiconductor device fabrication.

[1] Lithography: photolithography (mask aligner, maskless aligner), electron-beam lithography

[2] Thermal annealing: rapid thermal annealing and vacuum furnace

[3] Etching: reactive ion etching (RIE), inductively coupled plasma-RIE, Photo-enhanced metal-assisted chemical etching and wet chemical etching

[4] Metal deposition: electron-beam evaporator

[5] Oxidation of 2D semiconductors: UV/O_3 treatment

[6] Dry- and wet-transfer of 2/2.5D semiconductors

[7] Cleaning: wet cleaning (SPM, SC-1) and dry cleaning (UV/O_3, downstream plasma)

Ⅱ Device and material parameters analyze

[8] Electrical: I–V and C–V measurement

[9] Structural: AFM, SEM, EDS, EBSD

[10] Optical: PL and Raman spectroscopy

Ⅲ Software

[11] Ion implant simulation: SRIM

[12] Design: L-Edit, auto CAD

Teaching Experience

[1] 화학생물공정실험, 흡착실험, (2022. 1학기)

[2] 반도체 소재 및 공정, (2026. 1학기)